1.1 Complex Numbers and Their Properties

INTRODUCTION No one person "invented" complex numbers, but controversies surrounding the use of these numbers existed in the sixteenth century. In their quest to solve polynomial equations by formulas involving radicals, early dabblers in mathematics were forced to admit that there were other kinds of numbers besides positive integers. Equations such as $x^2 + 2x + 2 = 0$ and $x^3 = 6x + 4$ that yielded "solutions" $1 + \sqrt{-1}$ and $\sqrt[3]{2 + \sqrt{-2}} + \sqrt[3]{2 - \sqrt{-2}}$ caused particular consternation within the community of fledgling mathematical scholars because *everyone* knew that there are no numbers such as $\sqrt{-1}$ and $\sqrt{-2}$, numbers whose square is negative. Such "numbers" exist only in one's imagination, or as one philosopher opined, "the imaginary, (the) bosom child of complex mysticism." Over time these "imaginary numbers" did not go away, mainly because mathematicians as a group are tenacious and some are even practical. A famous mathematician held that even though "they exist in our imagination ... nothing prevents us from ... employing them in calculations." Mathematicians also hate to throw anything away. After all, a memory still lingered that negative numbers at first were branded "fictitious." The concept of *number* evolved over centuries; gradually the set of numbers grew from just positive integers to include rational numbers, negative numbers, and irrational numbers. But in the eighteenth century the number concept took a gigantic evolutionary step forward when the German mathematician Carl Friedrich Gauss put the so-called imaginary numbers—or *complex numbers*, as they were now beginning to be called—on a logical and consistent footing by treating them as an extension of the real number system.

Our goal in this first section is to examine some basic definitions and the arithmetic of complex numbers.

The Imaginary Unit

Even after gaining wide respectability, through the seminal works of Carl Friedrich Gauss and the French mathematician Augustin Louis Cauchy, the unfortunate name "imaginary" has survived down the centuries. The symbol i was originally used as a disguise for the embarrassing symbol $\sqrt{-1}$. We now say that i is the **imaginary unit** and define it by the property $i^2 = -1$. Using the imaginary unit, we build a general complex number out of two real numbers.

DEFINITION 1.1.1 Complex Number

A **complex number** is any number of the form $z = a + ib$ where a and b are real numbers and i is the imaginary unit.

Terminology

The notations $a + ib$ and $a + bi$ are used interchangeably. The real number a in $z = a + ib$ is called the **real part** of z; the real number b is called the **imaginary part** of z. The real and imaginary parts of a complex number z are abbreviated $\text{Re}(z)$ and $\text{Im}(z)$, respectively. For example, if $z = 4 - 9i$, then $\text{Re}(z) = 4$ and $\text{Im}(z) = -9$. A real constant multiple of the imaginary unit is called a **pure imaginary number**. For example, $z = 6i$ is a pure imaginary number. Two complex numbers are **equal** if their corresponding real and imaginary parts are equal. Since this simple concept is sometimes useful, we formalize the last statement in the next definition.

Note: The imaginary part of $z = 4 - 9i$ is -9 **not** $-9i$. ▶

DEFINITION 1.1.2 Equality

Complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are **equal**, $z_1 = z_2$, if $a_1 = a_2$ and $b_1 = b_2$.

In terms of the symbols $\text{Re}(z)$ and $\text{Im}(z)$, Definition 1.1.2 states that $z_1 = z_2$ if $\text{Re}(z_1) = \text{Re}(z_2)$ and $\text{Im}(z_1) = \text{Im}(z_2)$.

The totality of complex numbers or the set of complex numbers is usually denoted by the symbol $\mathbf{C}$. Because *any* real number a can be written as $z = a + 0i$, we see that the set $\mathbf{R}$ of real numbers is a subset of $\mathbf{C}$.

Arithmetic Operations Complex numbers can be added, subtracted, multiplied, and divided. If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, these operations are defined as follows.

Addition:
$$z_1 + z_2 = (a_1 + ib_1) + (a_2 + ib_2) = (a_1 + a_2) + i(b_1 + b_2)$$

Subtraction:
$$z_1 - z_2 = (a_1 + ib_1) - (a_2 + ib_2) = (a_1 - a_2) + i(b_1 - b_2)$$

Multiplication:
$$\begin{aligned} z_1 \cdot z_2 &= (a_1 + ib_1)(a_2 + ib_2) \\ &= a_1a_2 - b_1b_2 + i(b_1a_2 + a_1b_2) \end{aligned}$$

Division:
$$\begin{aligned} \frac{z_1}{z_2} &= \frac{a_1 + ib_1}{a_2 + ib_2}, \; a_2 \neq 0, \text{ or } b_2 \neq 0 \\ &= \frac{a_1a_2 + b_1b_2}{a_2^2 + b_2^2} + i\frac{b_1a_2 - a_1b_2}{a_2^2 + b_2^2} \end{aligned}$$

The familiar commutative, associative, and distributive laws hold for complex numbers:

Commutative laws:
$$\begin{cases} z_1 + z_2 = z_2 + z_1 \\ z_1z_2 = z_2z_1 \end{cases}$$

Associative laws:
$$\begin{cases} z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3 \\ z_1(z_2z_3) = (z_1z_2)z_3 \end{cases}$$

Distributive law: $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$

In view of these laws, there is no need to memorize the definitions of addition, subtraction, and multiplication.

Addition, Subtraction, and Multiplication

(*i*) *To add (subtract) two complex numbers, simply add (subtract) the corresponding real and imaginary parts.*

(*ii*) *To multiply two complex numbers, use the distributive law and the fact that* $i^2 = -1$.

The definition of division deserves further elaboration, and so we will discuss that operation in more detail shortly.

EXAMPLE 1 Addition and Multiplication

If $z_1 = 2 + 4i$ and $z_2 = -3 + 8i$, find (**a**) $z_1 + z_2$ and (**b**) $z_1 z_2$.

Solution (**a**) By adding real and imaginary parts, the sum of the two complex numbers z_1 and z_2 is

$$z_1 + z_2 = (2 + 4i) + (-3 + 8i) = (2 - 3) + (4 + 8)i = -1 + 12i.$$

(**b**) By the distributive law and $i^2 = -1$, the product of z_1 and z_2 is

$$\begin{aligned} z_1 z_2 = (2 + 4i)(-3 + 8i) &= (2 + 4i)(-3) + (2 + 4i)(8i) \\ &= -6 - 12i + 16i + 32i^2 \\ &= (-6 - 32) + (16 - 12)i = -38 + 4i. \end{aligned}$$

Zero and Unity The **zero** in the complex number system is the number $0+0i$ and the **unity** is $1+0i$. The zero and unity are denoted by 0 and 1, respectively. The zero is the **additive identity** in the complex number system since, for any complex number $z = a + ib$, we have $z + 0 = z$. To see this, we use the definition of addition:

$$z + 0 = (a + ib) + (0 + 0i) = a + 0 + i(b + 0) = a + ib = z.$$

Similarly, the unity is the **multiplicative identity** of the system since, for any complex number z, we have

$$z \cdot 1 = (a + ib) \cdot (1 + 0i) = a - 0 + i(b - 0) = a + ib = z.$$

Conjugate If z is a complex number, the number obtained by changing the sign of its imaginary part is called the **complex conjugate**, or simply **conjugate**, of z and is denoted by the symbol $\bar{z}$. In other words, if $z = a + ib$, then its conjugate is $\bar{z} = a - ib$. For example, if $z = 6 + 3i$, then $\bar{z} = 6 - 3i$; if $z = -5 - i$, then $\bar{z} = -5 + i$. If z is a real number, say, $z = 7$, then $\bar{z} = 7$. From the definitions of addition and subtraction of complex numbers, it is readily shown that the conjugate of a sum and difference of two complex numbers is the sum and difference of the conjugates:

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2. \tag{1}$$

Moreover, we have the following three additional properties:

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, \quad \bar{\bar{z}} = z. \tag{2}$$

Of course, the conjugate of any finite sum (product) of complex numbers is the sum (product) of the conjugates.

The definitions of addition and multiplication show that the sum and product of a complex number z with its conjugate $\bar{z}$ is a real number:

$$z + \bar{z} = (a + ib) + (a - ib) = 2a \tag{3}$$

$$z\bar{z} = (a + ib)(a - ib) = a^2 - i^2 b^2 = a^2 + b^2. \tag{4}$$

The difference of a complex number z with its conjugate $\bar{z}$ is a pure imaginary number:

$$z - \bar{z} = (a + ib) - (a - ib) = 2ib. \tag{5}$$

Since $a = \text{Re}(z)$ and $b = \text{Im}(z)$, (3) and (5) yield two useful formulas:

$$\text{Re}(z) = \frac{z + \bar{z}}{2} \quad \text{and} \quad \text{Im}(z) = \frac{z - \bar{z}}{2i}. \tag{6}$$

However, (4) is *the* important relationship in this discussion because it enables us to approach division in a practical manner.

Division

To divide z_1 by z_2, multiply the numerator and denominator of z_1/z_2 by the conjugate of z_2. That is,

$$\frac{z_1}{z_2} = \frac{z_1}{z_2} \cdot \frac{\bar{z}_2}{\bar{z}_2} = \frac{z_1\bar{z}_2}{z_2\bar{z}_2} \tag{7}$$

and then use the fact that $z_2\bar{z}_2$ is the sum of the squares of the real and imaginary parts of z_2.

The procedure described in (7) is illustrated in the next example.

EXAMPLE 2 Division

If $z_1 = 2 - 3i$ and $z_2 = 4 + 6i$, find z_1/z_2.

Solution We multiply numerator and denominator by the conjugate $\bar{z}_2 = 4 - 6i$ of the denominator $z_2 = 4 + 6i$ and then use (4):

$$\frac{z_1}{z_2} = \frac{2 - 3i}{4 + 6i} = \frac{2 - 3i}{4 + 6i}\frac{4 - 6i}{4 - 6i} = \frac{8 - 12i - 12i + 18i^2}{4^2 + 6^2} = \frac{-10 - 24i}{52}.$$

Because we want an answer in the form $a + bi$, we rewrite the last result by dividing the real and imaginary parts of the numerator $-10 - 24i$ by 52 and reducing to lowest terms:

$$\frac{z_1}{z_2} = -\frac{10}{52} - \frac{24}{52}i = -\frac{5}{26} - \frac{6}{13}i. \qquad \equiv$$

Inverses In the complex number system, every number z has a unique **additive inverse**. As in the real number system, the additive inverse of $z = a + ib$ is its *negative*, $-z$, where $-z = -a - ib$. For any complex number z, we have $z + (-z) = 0$. Similarly, every *nonzero* complex number z has a **multiplicative inverse**. In symbols, for $z \neq 0$ there exists one and only one nonzero complex number z^{-1} such that $zz^{-1} = 1$. The multiplicative inverse z^{-1} is the same as the **reciprocal** $1/z$.

EXAMPLE 3 Reciprocal

Find the reciprocal of $z = 2 - 3i$.

Solution By the definition of division we obtain

$$\frac{1}{z} = \frac{1}{2-3i} = \frac{1}{2-3i}\,\frac{2+3i}{2+3i} = \frac{2+3i}{4+9} = \frac{2+3i}{13}.$$

Answer should be in the form $a+ib$. ▶ That is,

$$\frac{1}{z} = z^{-1} = \frac{2}{13} + \frac{3}{13}i.$$

You should take a few seconds to verify the multiplication

$$zz^{-1} = (2-3i)\left(\tfrac{2}{13} + \tfrac{3}{13}i\right) = 1.$$

☰

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) Many of the properties of the real number system **R** hold in the complex number system **C**, but there are some truly remarkable differences as well. For example, the concept of *order* in the real number system does not carry over to the complex number system. In other words, we cannot compare two complex numbers $z_1 = a_1 + ib_1$, $b_1 \neq 0$, and $z_2 = a_2 + ib_2$, $b_2 \neq 0$, by means of inequalities. Statements such as $z_1 < z_2$ or $z_2 \geq z_1$ have no meaning in **C** except in the special case when the two numbers z_1 and z_2 are real. See Problem 55 in Exercises 1.1. Therefore, if you see a statement such as $z_1 = \alpha z_2$, $\alpha > 0$, it is implicit from the use of the inequality $\alpha > 0$ that the symbol α represents a real number.

(*ii*) Some things that we take for granted as impossible in real analysis, such as $e^x = -2$ and $\sin x = 5$ when x is a real variable, are perfectly correct and ordinary in complex analysis when the symbol x is interpreted as a complex variable. See Example 3 in Section 4.1 and Example 2 in Section 4.3.

We will continue to point out other differences between real analysis and complex analysis throughout the remainder of the text.

EXERCISES 1.1

Answers to selected odd-numbered problems begin on page ANS-1.

1. Evaluate the following powers of i.

(a) i^8 **(b)** i^{11}

(c) i^{42} **(d)** i^{105}

2. Write the given number in the form $a+ib$.

(a) $2i^3 - 3i^2 + 5i$ **(b)** $3i^5 - i^4 + 7i^3 - 10i^2 - 9$

(c) $\dfrac{5}{i} + \dfrac{2}{i^3} - \dfrac{20}{i^{18}}$ **(d)** $2i^6 + \left(\dfrac{2}{-i}\right)^3 + 5i^{-5} - 12i$

In Problems 3–20, write the given number in the form $a + ib$.

3. $(5 - 9i) + (2 - 4i)$

4. $3(4 - i) - 3(5 + 2i)$

5. $i(5 + 7i)$

6. $i(4 - i) + 4i(1 + 2i)$

7. $(2 - 3i)(4 + i)$

8. $\left(\frac{1}{2} - \frac{1}{4}i\right)\left(\frac{2}{3} + \frac{5}{3}i\right)$

9. $3i + \dfrac{1}{2 - i}$

10. $\dfrac{i}{1 + i}$

11. $\dfrac{2 - 4i}{3 + 5i}$

12. $\dfrac{10 - 5i}{6 + 2i}$

13. $\dfrac{(3 - i)(2 + 3i)}{1 + i}$

14. $\dfrac{(1 + i)(1 - 2i)}{(2 + i)(4 - 3i)}$

15. $\dfrac{(5 - 4i) - (3 + 7i)}{(4 + 2i) + (2 - 3i)}$

16. $\dfrac{(4 + 5i) + 2i^3}{(2 + i)^2}$

17. $i(1 - i)(2 - i)(2 + 6i)$

18. $(1 + i)^2(1 - i)^3$

19. $(3 + 6i) + (4 - i)(3 + 5i) + \dfrac{1}{2 - i}$

20. $(2 + 3i)\left(\dfrac{2 - i}{1 + 2i}\right)^2$

In Problems 21–24, use the **binomial theorem***

$$(A + B)^n = A^n + \frac{n}{1!}A^{n-1}B + \frac{n(n-1)}{2!}A^{n-2}B^2 + \cdots + \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}A^{n-k}B^k + \cdots + B^n,$$

where $n = 1,\ 2,\ 3,\ \ldots$, to write the given number in the form $a + ib$.

21. $(2 + 3i)^2$

22. $\left(1 - \frac{1}{2}i\right)^3$

23. $(-2 + 2i)^5$

24. $(1 + i)^8$

In Problems 25 and 26, find $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$.

25. $z = \left(\dfrac{i}{3 - i}\right)\left(\dfrac{1}{2 + 3i}\right)$

26. $z = \dfrac{1}{(1 + i)(1 - 2i)(1 + 3i)}$

In Problems 27–30, let $z = x + iy$. Express the given quantity in terms of x and y.

27. $\mathrm{Re}(1/z)$

28. $\mathrm{Re}(z^2)$

29. $\mathrm{Im}(2z + 4\bar{z} - 4i)$

30. $\mathrm{Im}(\bar{z}^2 + z^2)$

In Problems 31–34, let $z = x + iy$. Express the given quantity in terms of the symbols $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$.

31. $\mathrm{Re}(iz)$

32. $\mathrm{Im}(iz)$

33. $\mathrm{Im}((1 + i)z)$

34. $\mathrm{Re}(z^2)$

In Problems 35 and 36, show that the indicated numbers satisfy the given equation. In each case explain why additional solutions can be found.

35. $z^2 + i = 0$, $z_1 = -\dfrac{\sqrt{2}}{2} + \dfrac{\sqrt{2}}{2}i$. Find an additional solution, z_2.

36. $z^4 = -4$; $z_1 = 1 + i,\ z_2 = -1 + i$. Find two additional solutions, z_3 and z_4.

---

*Recall that the coefficients in the expansions of $(A + B)^2$, $(A + B)^3$, and so on, can also be obtained using Pascal's triangle.

In Problems 37–42, use Definition 1.1.2 to solve each equation for $z = a + ib$.

37. $2z = i(2 + 9i)$

38. $z - 2\bar{z} + 7 - 6i = 0$

39. $z^2 = i$

40. $\bar{z}^2 = 4z$

41. $z + 2\bar{z} = \dfrac{2 - i}{1 + 3i}$

42. $\dfrac{z}{1 + \bar{z}} = 3 + 4i$

In Problems 43 and 44, solve the given system of equations for z_1 and z_2.

43.
$$\begin{aligned} iz_1 - \quad & iz_2 = 2 + 10i \\ -z_1 + (1 - i)z_2 &= 3 - 5i \end{aligned}$$

44.
$$\begin{aligned} iz_1 + (1 + i)z_2 &= 1 + 2i \\ (2 - i)z_1 + \quad & 2iz_2 = 4i \end{aligned}$$

Focus on Concepts

45. What can be said about the complex number z if $z = \bar{z}$? If $(z)^2 = (\bar{z})^2$?

46. Think of an alternative solution to Problem 24. Then without doing any significant work, evaluate $(1 + i)^{5404}$.

47. For n a nonnegative integer, i^n can be one of four values: 1, i, -1, and $-i$. In each of the following four cases, express the integer exponent n in terms of the symbol k, where $k = 0, 1, 2, \ldots$.

(**a**) $i^n = 1$

(**b**) $i^n = i$

(**c**) $i^n = -1$

(**d**) $i^n = -i$

48. There is an alternative to the procedure given in (7). For example, the quotient $(5 + 6i)/(1 + i)$ must be expressible in the form $a + ib$:

$$\frac{5 + 6i}{1 + i} = a + ib.$$

Therefore, $5 + 6i = (1 + i)(a + ib)$. Use this last result and Definition 1.1.2 to find the given quotient. Use this method to find the reciprocal of $3 - 4i$.

49. Assume for the moment that $\sqrt{1 + i}$ makes sense in the complex number system. How would you then demonstrate the validity of the equality

$$\sqrt{1 + i} = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{2}} + i\sqrt{-\tfrac{1}{2} + \tfrac{1}{2}\sqrt{2}}?$$

50. Suppose that z_1 and z_2 are complex numbers. Use Definition 1.1.2 to prove that if $z_1 z_2 = 0$, then $z_1 = 0$ or $z_2 = 0$.

51. Suppose the product $z_1 z_2$ of two complex numbers is a nonzero real constant. Show that $z_2 = k\bar{z}_1$, where k is a real number.

52. Without doing any significant work, explain why it follows immediately from (2) and (3) that $z_1\bar{z}_2 + \bar{z}_1 z_2 = 2\text{Re}(z_1\bar{z}_2)$.

53. Mathematicians like to prove that certain "things" within a mathematical system are unique. For example, a proof of a proposition such as "The unity in the complex number system is unique" usually starts out with the assumption that there exist two *different* unities, say, 1_1 and 1_2, and then proceeds to show that this assumption leads to some contradiction. Give one contradiction if it is assumed that two different unities exist.

54. Follow the procedure outlined in Problem 53 to prove the proposition "The zero in the complex number system is unique."

55. A number system is said to be an **ordered system** provided it contains a subset P with the following two properties:

First, for any nonzero number x in the system, either x or $-x$ is (but not both) in P.

Second, if x and y are numbers in P, then both xy and $x + y$ are in P.

In the real number system the set P is the set of *positive* numbers. In the real number system we say x is greater than y, written $x > y$, if and only if $x - y$ is in P. Discuss why the complex number system has no such subset P. [*Hint*: Consider i and $-i$.]

Computer Lab Assignments

In Problems 56–59, use a CAS* to write the given complex number in the form $a + bi$. In *Mathematica* use the command **ComplexExpand**.

56. $2 + i - \dfrac{1}{3 - i}$ **57.** $\dfrac{5i}{2 + 3i}$

58. $\dfrac{1 + i}{2 - 5i}$ **59.** $\dfrac{(1 + 3i)(2 + i)}{3 - i}$

1.2 Complex Plane

INTRODUCTION A complex number $z = x + iy$ is uniquely determined by an *ordered pair* of real numbers (x, y). The first and second entries of the ordered pairs correspond, in turn, with the real and imaginary parts of the complex number. For example, the ordered pair $(2, -3)$ corresponds to the complex number $z = 2 - 3i$. Conversely, $z = 2 - 3i$ determines the ordered pair $(2, -3)$. The numbers 7, i, and $-5i$ are equivalent to $(7, 0)$, $(0, 1)$, $(0, -5)$, respectively. In this manner we are able to associate a complex number $z = x + iy$ with a *point* (x, y) in a coordinate plane.

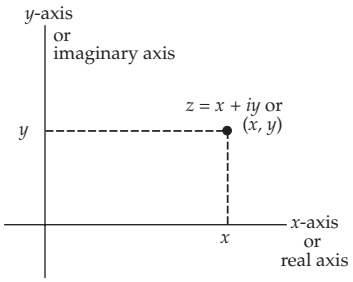


FIGURE 1.2.1 z-plane

Complex Plane Because of the correspondence between a complex number $z = x + iy$ and one and only one point (x, y) in a coordinate plane, we shall use the terms *complex number* and *point* interchangeably. The coordinate plane illustrated in FIGURE 1.2.1 is called the **complex plane** or simply the **z-plane**. The horizontal or x-axis is called the **real axis** because each point on that axis represents a real number. The vertical or y-axis is called the **imaginary axis** because a point on that axis represents a pure imaginary number.

Vectors In other courses you have undoubtedly seen that the numbers in an ordered pair of real numbers can be interpreted as the components of a vector. Thus, a complex number $z = x + iy$ can also be viewed as a two-dimensional position **vector**, that is, a vector whose initial point is the origin and whose terminal point is the point (x, y). See FIGURE 1.2.2. This vector interpretation prompts us to define the length of the vector z as the distance $\sqrt{x^2 + y^2}$ from the origin to the point (x, y). This length is given a special name.

A brief review of vectors can be found in Section 1.2 of the student study guide. ▶

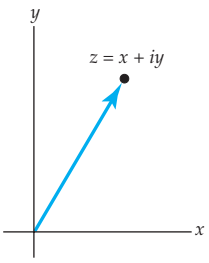


FIGURE 1.2.2 z as a position vector

DEFINITION 1.2.1 Modulus

The **modulus** of a complex number $z = x + iy$ is the real number

$$|z| = \sqrt{x^2 + y^2}. \tag{1}$$

The modulus $|z|$ of a complex number z is also called the **absolute value** of z. We shall use both words *modulus* and *absolute value* throughout this text.

EXAMPLE 1 Modulus of a Complex Number

If $z = 2 - 3i$, then from (1) we find the modulus of the number to be $|z| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$. If $z = -9i$, then (1) gives $|-9i| = \sqrt{(-9)^2} = 9$. ☰

*Throughout this text we shall use the abbreviation CAS for "computer algebra system."

Properties Recall from (4) of Section 1.1 that for any complex number $z = x + iy$ the product $z\bar{z}$ is a real number; specifically, $z\bar{z}$ is the sum of the squares of the real and imaginary parts of z: $z\bar{z} = x^2 + y^2$. Inspection of (1) then shows that $|z|^2 = x^2 + y^2$. The relations

$$|z|^2 = z\bar{z} \quad \text{and} \quad |z| = \sqrt{z\bar{z}} \tag{2}$$

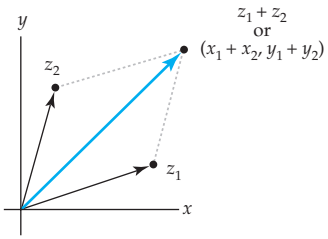
deserve to be stored in memory. The modulus of a complex number z has the additional properties.

$$|z_1 z_2| = |z_1|\,|z_2| \quad \text{and} \quad \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}. \tag{3}$$

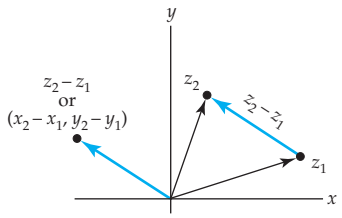
Note that when $z_1 = z_2 = z$, the first property in (3) shows that

$$|z^2| = |z|^2. \tag{4}$$

The property $|z_1 z_2| = |z_1|\,|z_2|$ can be proved using (2) and is left as an exercise. See Problem 49 in Exercises 1.2.



(a) Vector sum



(b) Vector difference

FIGURE 1.2.3 Sum and difference of vectors

Distance Again The addition of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ given in Section 1.1, when stated in terms of ordered pairs:

$$(x_1,\ y_1) + (x_2,\ y_2) = (x_1 + x_2,\ y_1 + y_2)$$

is simply the component definition of vector addition. The vector interpretation of the sum $z_1 + z_2$ is the vector shown in **FIGURE 1.2.3(a)** as the main diagonal of a parallelogram whose initial point is the origin and terminal point is $(x_1 + x_2,\ y_1 + y_2)$. The difference $z_2 - z_1$ can be drawn either starting from the terminal point of z_1 and ending at the terminal point of z_2, or as a position vector whose initial point is the origin and terminal point is $(x_2 - x_1,\ y_2 - y_1)$. See Figure 1.2.3(b). In the case $z = z_2 - z_1$, it follows from (1) and Figure 1.2.3(b) that the **distance between two points** $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ in the complex plane is the same as the distance between the origin and the point $(x_2 - x_1,\ y_2 - y_1)$; that is, $|z| = |z_2 - z_1| = |(x_2 - x_1) + i(y_2 - y_1)|$ or

$$|z_2 - z_1| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \tag{5}$$

When $z_1 = 0$, we see again that the modulus $|z_2|$ represents the distance between the origin and the point z_2.

EXAMPLE 2 Set of Points in the Complex Plane

Describe the set of points z in the complex plane that satisfy $|z| = |z - i|$.

Solution We can interpret the given equation as equality of distances: The distance from a point z to the origin equals the distance from z to the point i. Geometrically, it seems plausible from **FIGURE 1.2.4** that the set of points z lie on a horizontal line. To establish this analytically, we use (1) and (5) to write $|z| = |z - i|$ as:

$$\sqrt{x^2 + y^2} = \sqrt{x^2 + (y - 1)^2}$$
$$x^2 + y^2 = x^2 + (y - 1)^2$$
$$x^2 + y^2 = x^2 + y^2 - 2y + 1.$$

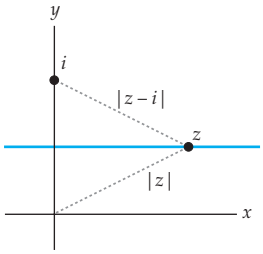


FIGURE 1.2.4 Horizontal line is the set of points satisfying $|z| = |z - i|$

The last equation yields $y = \frac{1}{2}$. Since the equality is true for arbitrary x, $y = \frac{1}{2}$ is an equation of the horizontal line shown in color in Figure 1.2.4. Complex numbers satisfying $|z| = |z - i|$ can then be written as $z = x + \frac{1}{2}i$. ☰

Inequalities In the *Remarks* at the end of the last section we pointed out that no order relation can be defined on the system of complex numbers. However, since $|z|$ is a real number, we can compare the absolute values of two complex numbers. For example, if $z_1 = 3 + 4i$ and $z_2 = 5 - i$, then $|z_1| = \sqrt{25} = 5$ and $|z_2| = \sqrt{26}$ and, consequently, $|z_1| < |z_2|$. In view of (1), a geometric interpretation of the last inequality is simple: *The point* (3, 4) *is closer to the origin than the point* (5, −1).

Now consider the triangle given in **FIGURE 1.2.5** with vertices at the origin, z_1, and $z_1 + z_2$. We know from geometry that the length of the side of the triangle corresponding to the vector $z_1 + z_2$ cannot be longer than the sum of the lengths of the remaining two sides. In symbols we can express this observation by the inequality

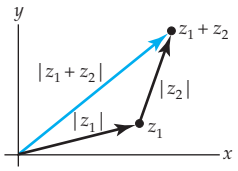


FIGURE 1.2.5 Triangle with vector sides

$$|z_1 + z_2| \le |z_1| + |z_2|. \tag{6}$$

This inequality can be derived using the properties of complex numbers in Section 1.1. See Problem 50 in Exercises 1.2. ▶

The result in (6) is known as the **triangle inequality**. Now from the identity $z_1 = z_1 + z_2 + (-z_2)$, (6) gives

$$|z_1| = |z_1 + z_2 + (-z_2)| \le |z_1 + z_2| + |-z_2|\,.$$

Since $|z_2| = |-z_2|$ (see Problem 47 in Exercises 1.2), solving the last result for $|z_1 + z_2|$ yields another important inequality:

$$|z_1| - |z_2| \le |z_1 + z_2|. \tag{7}$$

But because $z_1 + z_2 = z_2 + z_1$, (7) can be written in the alternative form $|z_1 + z_2| = |z_2 + z_1| \ge |z_2| - |z_1| = -(|z_1| - |z_2|)$ and so combined with the last result implies

$$\big|\,|z_1| - |z_2|\,\big| \le |z_1 + z_2|. \tag{8}$$

It also follows from (6) by replacing z_2 by $-z_2$ that $|z_1 + (-z_2)| \le |z_1| + |(-z_2)| = |z_1| + |z_2|$. This result is the same as

$$|z_1 - z_2| \le |z_1| + |z_2|\,. \tag{9}$$

From (8) with z_2 replaced by $-z_2$, we also find

$$\big|\,|z_1| - |z_2|\,\big| \le |z_1 - z_2|. \tag{10}$$

In conclusion, we note that the triangle inequality (6) extends to any finite sum of complex numbers:

$$|z_1 + z_2 + z_3 + \cdots + z_n| \le |z_1| + |z_2| + |z_3| + \cdots + |z_n|\,. \tag{11}$$

The inequalities (6), (8), and (10) will be important when we work with integrals in Chapters 5 and 6.

EXAMPLE 3 An Upper Bound

Find an upper bound for $\left|\dfrac{-1}{z^4+3z^2+2}\right|$ if $|z|=2$.

Solution By the second result in (3), the absolute value of a quotient is the quotient of the absolute values. Thus with $|-1|=1$, we want to find a positive real number M such that

$$\frac{1}{|z^4+3z^2+2|} \le M.$$

To accomplish this task we want the denominator as small as possible. Since $z^4+3z^2+2=(z^2+1)(z^2+2)$ we can use the first result in (3) and (8) to write

$$|z^4+3z^2+2| = |z^2+1|\cdot|z^2+2| \ge \big||z^2|-1\big|\cdot\big||z^2|-2\big|. \tag{12}$$

Using $|z|=2$, (12) becomes

$$\begin{aligned} |z^4+3z^2+2| \ge \big||z^2|-1\big|\cdot\big||z^2|-2\big| &= \big||z|^2-1\big|\cdot\big||z|^2-2\big| \\ &= |4-1|\cdot|4-2| \\ &= 6. \end{aligned}$$

Hence for $|z|=2$ we have

$$\left|\frac{-1}{z^4+3z^2+2}\right| = \frac{1}{|z^4+3z^2+2|} \le \frac{1}{6}.$$

☰

Remarks

We have seen that the triangle inequality $|z_1+z_2| \le |z_1|+|z_2|$ indicates that the length of the vector z_1+z_2 cannot exceed the sum of the lengths of the individual vectors z_1 and z_2. But the results given in (3) are interesting. The product z_1z_2 and quotient z_1/z_2, $(z_2 \neq 0)$, are complex numbers and so are vectors in the complex plane. The equalities $|z_1z_2| = |z_1|\,|z_2|$ and $|z_1/z_2| = |z_1|/|z_2|$ indicate that the lengths of the vectors z_1z_2 and z_1/z_2 are exactly equal to the product of the lengths and to the quotient of the lengths, respectively, of the individual vectors z_1 and z_2.

EXERCISES 1.2

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–4, interpret z_1 and z_2 as vectors. Graph z_1, z_2, and the indicated sum and difference as vectors.

1. $z_1 = 4+2i,\ z_2 = -2+5i;\ z_1+z_2,\ z_1-z_2$
2. $z_1 = 1-i,\ z_2 = 1+i;\ z_1+z_2,\ z_1-z_2$
3. $z_1 = 5+4i,\ z_2 = -3i;\ 3z_1+5z_2,\ z_1-2z_2$
4. $z_1 = 4-3i,\ z_2 = -2+3i;\ 2z_1+4z_2,\ z_1-z_2$
5. Given that $z_1 = 5-2i$ and $z_2 = -1-i$, find a vector z_3 in the same direction as z_1+z_2 but four times as long.

6. (**a**) Plot the points $z_1 = -2 - 8i,\ z_2 = 3i,\ z_3 = -6 - 5i$.

(**b**) The points in part (a) determine a triangle with vertices at z_1, z_2, and z_3, respectively. Express each side of the triangle as a difference of vectors.

7. In Problem 6, determine whether the points z_1, z_2, and z_3 are the vertices of a right triangle.

8. The three points $z_1 = 1 + 5i,\ z_2 = -4 - i,\ z_3 = 3 + i$ are vertices of a triangle. Find the length of the median from z_1 to the side $z_3 - z_2$.

In Problems 9–12, find the modulus of the given complex number.

9. $(1 - i)^2$

10. $i(2 - i) - 4(1 + \frac{1}{4}i)$

11. $\dfrac{2i}{3 - 4i}$

12. $\dfrac{1 - 2i}{1 + i} + \dfrac{2 - i}{1 - i}$

In Problems 13 and 14, let $z = x + iy$. Express the given quantity in terms of x and y.

13. $|z - 1 - 3i|^2$

14. $|z + 5\bar{z}|$

In Problems 15 and 16, determine which of the given two complex numbers is closest to the origin. Which is closest to $1 + i$?

15. $10 + 8i,\ 11 - 6i$

16. $\frac{1}{2} - \frac{1}{4}i,\ \frac{2}{3} + \frac{1}{6}i$

In Problems 17–26, describe the set of points z in the complex plane that satisfy the given equation. Include a plot of the set of points in the complex plane.

17. $\text{Re}((1 + i)z - 1) = 0$

18. $[\text{Im}(i\bar{z})]^2 = 2$

19. $|z - i| = |z - 1|$

20. $\bar{z} = z^{-1}$

21. $\text{Im}(z^2) = 2$

22. $\text{Re}(z^2) = \left|\sqrt{3} - i\right|$

23. $|z - 1| = 1$

24. $|z - i| = 2\,|z - 1|$

25. $|z - 2| = \text{Re}(z)$

26. $|z| = \text{Re}(z)$

In Problems 27 and 28, establish the given simultaneous inequality.

27. If $|z| = 2$, then $8 \le |\,z + 6 + 8i\,| \le 12$.

28. If $|z| = 1$, then $2 \le \left|\,z^2 - 3\,\right| \le 4$.

29. Find an upper bound for the modulus of $3z^2 + 2z + 1$ if $|z| \le 1$.

30. Find an upper and lower bound for the reciprocal of the modulus of $z^4 - 5z^2 + 6$ if $|z| = 2$. [*Hint*: $z^4 - 5z^2 + 6 = (z^2 - 3)(z^2 - 2)$.]

In Problems 31 and 32, find a number z that satisfies the given equation.

31. $|z| - z = 2 + i$

32. $|z|^2 + 1 + 12i = 6z$

Focus on Concepts

33. (**a**) Draw the pair of points $z = a + ib$ and $\bar{z} = a - ib$ in the complex plane if

$$a > 0, b > 0; a > 0, b < 0; a < 0, b > 0; \text{and} a < 0,\ b < 0.$$

(**b**) In general, how would you describe geometrically the relationship between a complex number $z = a + ib$ and its conjugate $\bar{z} = a - ib$?

(**c**) Describe geometrically the relationship between $z = a + ib$ and $z_1 = -a + ib$.

34. How would you describe geometrically the relationship between a nonzero complex number $z = a + ib$ and its

(**a**) negative, $-z$?

(**b**) inverse, z^{-1}? [*Hint*: Reread Problem 33 and then recall $z^{-1} = \bar{z}/|z|^2$.]

35. Consider the complex numbers $z_1 = 4 + i$, $z_2 = -2 + i$, $z_3 = -2 - 2i$, $z_4 = 3 - 5i$.

(**a**) Use four different sketches to plot the four pairs of points z_1, iz_1; z_2, iz_2; z_3, iz_3; and z_4, iz_4.

(**b**) In general, how would you describe geometrically the effect of multiplying a complex number $z = x + iy$ by i? By $-i$?

36. Use Definition 1.1.2 to prove that $|z| = 0$ if and only if $z = 0$.

37. Under what circumstances does $|z_1 + z_2| = |z_1| + |z_2|$?

38. Using the complex variable z, find an equation of a circle in the complex plane of radius 5 centered at $3 - 6i$. [*Hint*: Use (5).]

39. Describe the set of points z in the complex plane that satisfy $z = \cos\theta + i\sin\theta$, where θ is measured in radians from the positive x-axis.

40. Using the complex variable z, find an equation of an ellipse in the complex plane with foci $-2 + i$ and $2 + i$ whose major axis is 8 units long.

41. Suppose $z = x + iy$. In (6) of Section 1.1 we saw that x and y could be expressed in terms of z and $\bar{z}$. Use these results to express the following Cartesian equations in complex form.

(**a**) $x = 3$ (**b**) $y = 10$

(**c**) $y = x$ (**d**) $x + 2y = 8$

42. Using complex notation, find a parametric equation of the line segment between any two distinct complex numbers z_1 and z_2 in the complex plane.

43. Suppose z_1, z_2, and z_3 are three distinct points in the complex plane and k is a real number. Interpret $z_3 - z_2 = k(z_2 - z_1)$ geometrically.

44. Suppose $z_1 \neq z_2$. Interpret $\text{Re}(z_1\bar{z}_2) = 0$ geometrically in terms of vectors z_1 and z_2.

45. Suppose $w = \bar{z}/z$. Without doing any calculations, explain why $|w| = 1$.

46. Without doing any calculations, explain why the inequalities $|\text{Re}(z)| \leq |z|$ and $|\text{Im}(z)| \leq |z|$ hold for all complex numbers z.

47. Show that

(**a**) $|z| = |-z|$ (**b**) $|z| = |\bar{z}|$.

48. For any two complex numbers z_1 and z_2, show that

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2).$$

49. In this problem we will start you out in the proof of the first property $|z_1 z_2| = |z_1|\,|z_2|$ in (3). By the first result in (2) we can write $|z_1 z_2|^2 = (z_1 z_2)\overline{(z_1 z_2)}$. Now use the first property in (2) of Section 1.1 to continue the proof.

50. In this problem we guide you through an analytical proof of the triangle inequality (6).

Since $|z_1 + z_2|$ and $|z_1| + |z_2|$ are positive real numbers, we have

$|z_1 + z_2| \leq |z_1| + |z_2|$ if and only if $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$. Thus, it suffices to show that $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$.

(**a**) Explain why $|z_1 + z_2|^2 = |z_1|^2 + 2\text{Re}(z_1\bar{z}_2) + |z_2|^2$.

(**b**) Explain why $(|z_1| + |z_2|)^2 = |z_1|^2 + 2\,|z_1\bar{z}_2| + |z_2|^2$.

(**c**) Use parts (a) and (b) along with the results in Problem 46 to derive (6).

1.3 Polar Form of Complex Numbers

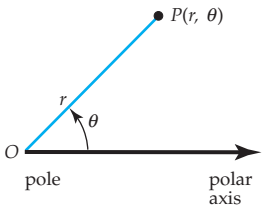


FIGURE 1.3.1 Polar coordinates

INTRODUCTION Recall from calculus that a point P in the plane whose rectangular coordinates are (x, y) can also be described in terms of **polar coordinates**. The polar coordinate system, invented by Isaac Newton, consists of point O called the **pole** and the horizontal half-line emanating from the pole called the **polar axis**. If r is a directed distance from the pole to P and θ is an angle of inclination (in radians) measured from the polar axis to the line OP, then the point can be described by the ordered pair (r, θ), called the polar coordinates of P. See FIGURE 1.3.1.

A brief review of polar coordinates can be found in Section 1.3 of the student study guide. ▶

Polar Form Suppose, as shown in FIGURE 1.3.2, that a polar coordinate system is superimposed on the complex plane with the polar axis coinciding with the positive x-axis and the pole O at the origin. Then x, y, r, and θ are related by $x = r\cos\theta$, $y = r\sin\theta$. These equations enable us to express a nonzero complex number $z = x + iy$ as $z = (r\cos\theta) + i(r\sin\theta)$ or

$$z = r(\cos\theta + i\sin\theta). \tag{1}$$

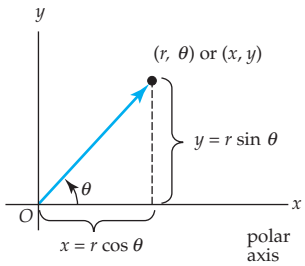


FIGURE 1.3.2 Polar coordinates in the complex plane

We say that (1) is the **polar form** or **polar representation** of the complex number z. Again, from Figure 1.3.2 we see that the coordinate r can be interpreted as the distance from the origin to the point (x, y). In other words, we shall adopt the convention that *r is never negative** so that we can take r to be the modulus of z, that is, $r = |z|$. The angle θ of inclination of the vector z, which *will always be measured in radians* from the positive real axis, is positive when measured counterclockwise and negative when measured clockwise. The angle θ is called an **argument** of z and is denoted by $\theta = \arg(z)$. An argument θ of a complex number must satisfy the equations $\cos\theta = x/r$ and $\sin\theta = y/r$. An argument of a complex number z is not unique since $\cos\theta$ and $\sin\theta$ are 2π-periodic; in other words, if θ_0 is an argument of z, then necessarily the angles $\theta_0 \pm 2\pi$, $\theta_0 \pm 4\pi$, ... are also arguments of z. In practice we use $\tan\theta = y/x$ to find θ. However, because $\tan\theta$ is π-periodic, some care must be exercised in using the last equation. A calculator will give only angles satisfying $-\pi/2 < \tan^{-1}(y/x) < \pi/2$, that is, angles in the first and fourth quadrants. We have to choose θ consistent with the quadrant in which z is located; this may require adding or subtracting π to $\tan^{-1}(y/x)$ when appropriate. The following example illustrates how this is done.

Be careful using $\tan^{-1}(y/x)$ ▶

EXAMPLE 1 A Complex Number in Polar Form

Express $-\sqrt{3} - i$ in polar form.

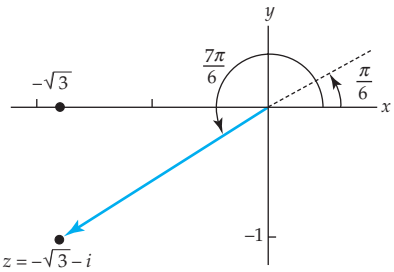


FIGURE 1.3.3 $\arg(-\sqrt{3} - i)$

Solution With $x = -\sqrt{3}$ and $y = -1$ we obtain $r = |z| = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$. Now $y/x = -1/(-\sqrt{3}) = 1/\sqrt{3}$, and so a calculator gives $\tan^{-1}(1/\sqrt{3}) = \pi/6$, which is an angle whose terminal side is in the first quadrant. But since the point $-\sqrt{3} - i$ lies in the third quadrant, we take the solution of $\tan\theta = -1/(-\sqrt{3}) = 1/\sqrt{3}$ to be $\theta = \arg(z) = \pi/6 + \pi = 7\pi/6$. See FIGURE 1.3.3. It follows from (1) that a polar form of the number is

$$z = 2\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right). \tag{2}$$

☰

*In general, in the polar description (r, θ) of a point P in the Cartesian plane, we can have $r \geq 0$ or $r < 0$.

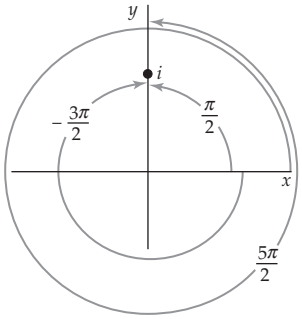


FIGURE 1.3.4 Some arguments of i

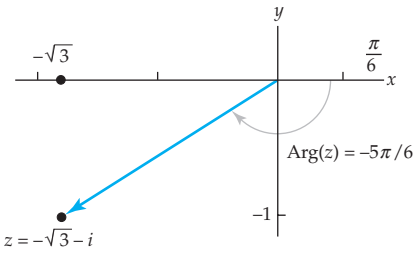


FIGURE 1.3.5 Principal argument $z = -\sqrt{3} - i$

Principal Argument The symbol $\arg(z)$ represents a set of values, but the choice of an argument θ of a complex number that lies in the interval $-\pi < \theta \leq \pi$ is called the **principal value** of $\arg(z)$ or the **principal argument** of z. The principal argument of z is *unique* and is represented by the symbol $\text{Arg}(z)$, that is,

$$-\pi < \text{Arg}(z) \leq \pi.$$

For example, if $z = i$, we see in **FIGURE 1.3.4** that some values of $\arg(i)$ are $\pi/2,\ 5\pi/2,\ -3\pi/2$, and so on, but $\text{Arg}(i) = \pi/2$. Similarly, we see from **FIGURE 1.3.5** that the argument of $-\sqrt{3} - i$ that lies in the interval $(-\pi,\ \pi]$, the principal argument of z, is $\text{Arg}(z) = \pi/6 - \pi = -5\pi/6$. Using $\text{Arg}(z)$ we can express the complex number in (2) in the alternative polar form:

$$z = 2\left[\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right].$$

In general, $\arg(z)$ and $\text{Arg}(z)$ are related by

$$\arg(z) = \text{Arg}(z) + 2n\pi, \quad n = 0,\ \pm 1,\ \pm 2, \ldots. \tag{3}$$

For example, $\arg(i) = \dfrac{\pi}{2} + 2n\pi$. For the choices $n = 0$ and $n = -1$, (3) gives $\arg(i) = \text{Arg}(i) = \pi/2$ and $\arg(i) = -3\pi/2$, respectively.

Multiplication and Division The polar form of a complex number is especially convenient when multiplying or dividing two complex numbers. Suppose

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1) \quad \text{and} \quad z_2 = r_2(\cos\theta_2 + i\sin\theta_2),$$

where θ_1 and θ_2 are any arguments of z_1 and z_2, respectively. Then

$$z_1 z_2 = r_1 r_2\left[\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 + i\left(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2\right)\right] \tag{4}$$

and, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\left[\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2 + i\left(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2\right)\right]. \tag{5}$$

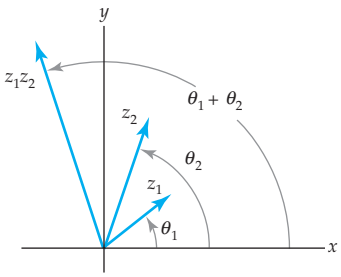


FIGURE 1.3.6 $\arg(z_1z_2) = \theta_1 + \theta_2$

From the addition formulas* for the cosine and sine, (4) and (5) can be rewritten as

$$z_1z_2 = r_1r_2\,[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)] \tag{6}$$

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\,[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]. \tag{7}$$

Inspection of the expressions in (6) and (7) and **FIGURE 1.3.6** shows that the lengths of the two vectors z_1z_2 and z_1/z_2 are the product of the lengths of z_1 and z_2 and the quotient of the lengths of z_1 and z_2, respectively. See (3) of Section 1.2. Moreover, the arguments of z_1z_2 and z_1/z_2 are given by

$$\arg(z_1z_2) = \arg(z_1) + \arg(z_2) \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2). \tag{8}$$

EXAMPLE 2 Argument of a Product and of a Quotient

We have just seen that for $z_1 = i$ and $z_2 = -\sqrt{3} - i$ that $\text{Arg}(z_1) = \pi/2$ and $\text{Arg}(z_2) = -5\pi/6$, respectively. Thus arguments for the product and quotient

$$z_1z_2 = i(-\sqrt{3} - i) = 1 - \sqrt{3}i \quad \text{and} \quad \frac{z_1}{z_2} = \frac{i}{-\sqrt{3} - i} = -\frac{1}{4} - \frac{\sqrt{3}}{4}i$$

can be obtained from (8):

$$\arg(z_1z_2) = \frac{\pi}{2} + \left(-\frac{5\pi}{6}\right) = -\frac{\pi}{3} \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2} - \left(-\frac{5\pi}{6}\right) = \frac{4\pi}{3}.$$

Additional values of $\arg(z_1z_2)$ and $\arg(z_1/z_2)$ can be found by adding integer multiples of 2π to $= \pi/3$ or $4\pi/3$, respectively.

☰

Integer Powers of z We can find integer powers of a complex number z from the results in (6) and (7). For example, if $z = r(\cos\theta + i\sin\theta)$, then with $z_1 = z_2 = z$, (6) gives

$$z^2 = r^2\,[\cos(\theta + \theta) + i\sin(\theta + \theta)] = r^2\,(\cos 2\theta + i\sin 2\theta)\,.$$

Since $z^3 = z^2z$, it then follows that

$$z^3 = r^3\,(\cos 3\theta + i\sin 3\theta)\,,$$

and so on. In addition, if we take $\arg(1) = 0$, then (7) gives

$$z^{-1} = \frac{1}{z} = \frac{1}{r}\,[\cos(0 - \theta) + i\,\sin(0 - \theta)] = r^{-1}\,[\cos(-\theta) + i\sin(-\theta)]\,.$$

Continuing in this manner, we obtain a formula for the nth power of z for any integer n:

$$z^n = r^n\,(\cos n\theta + i\sin n\theta). \tag{9}$$

When $n = 0$, we get the familiar result $z^0 = 1$.

---

*$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$ and $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$

EXAMPLE 3 Power of a Complex Number

Compute z^3 for $z = -\sqrt{3} - i$.

Solution In (2) of Example 1 we saw that a polar form of the given number is $z = 2[\cos(7\pi/6) + i\sin(7\pi/6)]$. Using (9) with $r = 2$, $\theta = 7\pi/6$, and $n = 3$ we get

$$\left(-\sqrt{3} - i\right)^3 = 2^3\left[\cos\left(3\frac{7\pi}{6}\right) + i\sin\left(3\frac{7\pi}{6}\right)\right] = 8\left[\cos\frac{7\pi}{2} + i\sin\frac{7\pi}{2}\right] = -8i$$

since $\cos(7\pi/2) = 0$ and $\sin(7\pi/2) = -1$. ☰

Note in Example 3, if we also want the value of z^{-3}, then we could proceed in two ways: either find the reciprocal of $z^3 = -8i$ or use (9) with $n = -3$.

de Moivre's Formula When $z = \cos\theta + i\sin\theta$, we have $|z| = r = 1$, and so (9) yields

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta. \tag{10}$$

This last result is known as **de Moivre's formula** and is useful in deriving certain trigonometric identities involving $\cos n\theta$ and $\sin n\theta$. See Problems 33 and 34 in Exercises 1.3.

EXAMPLE 4 de Moivre's Formula

From (10), with $\theta = \pi/6$, $\cos\theta = \sqrt{3}/2$, and $\sin\theta = 1/2$:

$$\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^3 = \cos 3\theta + i\sin 3\theta = \cos\left(3\cdot\frac{\pi}{6}\right) + i\sin\left(3\cdot\frac{\pi}{6}\right)$$
$$= \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i.$$

☰

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) Observe in Example 2 that even though we used the principal arguments of z_1 and z_2 that $\arg(z_1/z_2) = 4\pi/3 \neq \text{Arg}(z_1/z_2)$. Although (8) is true for any arguments of z_1 and z_2, *it is not true*, in general, that $\text{Arg}(z_1z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$ and $\text{Arg}(z_1/z_2) = \text{Arg}(z_1) - \text{Arg}(z_2)$. See Problems 37 and 38 in Exercises 1.3.

(*ii*) An argument can be assigned to any *nonzero* complex number z. However, for $z = 0$, $\arg(z)$ cannot be defined in any way that is meaningful.

(*iii*) If we take $\arg(z)$ from the interval $(-\pi,\ \pi]$, the relationship between a complex number z and its argument is single-valued; that is, every nonzero complex number has precisely *one* angle in $(-\pi,\ \pi]$. But there is nothing special about the interval $(-\pi,\ \pi]$; we also establish a single-valued relationship by using the interval $(0,\ 2\pi]$ to define the principal value of the argument of z. For the interval $(-\pi,\ \pi]$, the negative real axis is analogous to a barrier that we agree not to cross; the technical name for this barrier is a **branch cut**. If we use $(0,\ 2\pi]$, the branch cut is the positive real axis. The concept of a branch cut is important and will be examined in greater detail when we study functions in Chapters 3 and 4.

(*iv*) The "cosine *i* sine" part of the polar form of a complex number is sometimes abbreviated cis. That is,

$$z = r(\cos\theta + i\sin\theta) = r\,\text{cis}\,\theta.$$

This notation, used mainly in engineering, will not be used in this text.

EXERCISES 1.3

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–10, write the given complex number in polar form first using an argument $\theta \neq \text{Arg}(z)$ and then using $\theta = \text{Arg}(z)$.

1. 2

2. -10

3. $-3i$

4. $6i$

5. $1 + i$

6. $5 - 5i$

7. $-\sqrt{3} + i$

8. $-2 - 2\sqrt{3}i$

9. $\dfrac{3}{-1+i}$

10. $\dfrac{12}{\sqrt{3}+i}$

In Problems 11 and 12, use a calculator to write the given complex number in polar form first using an argument $\theta \neq \text{Arg}(z)$ and then using $\theta = \text{Arg}(z)$.

11. $-\sqrt{2} + \sqrt{7}i$

12. $-12 - 5i$

In Problems 13 and 14, write the complex number whose polar coordinates $(r,\ \theta)$ are given in the form $a + ib$. Use a calculator if necessary.

13. $(4,\ -5\pi/3)$

14. $(2,\ 2)$

In Problems 15–18, write the complex number whose polar form is given in the form $a + ib$. Use a calculator if necessary.

15. $z = 5\left(\cos\dfrac{7\pi}{6} + i\sin\dfrac{7\pi}{6}\right)$

16. $z = 8\sqrt{2}\left(\cos\dfrac{11\pi}{4} + i\sin\dfrac{11\pi}{4}\right)$

17. $z = 6\left(\cos\dfrac{\pi}{8} + i\sin\dfrac{\pi}{8}\right)$

18. $z = 10\left(\cos\dfrac{\pi}{5} + i\sin\dfrac{\pi}{5}\right)$

In Problems 19 and 20, use (6) and (7) to find z_1z_2 and z_1/z_2. Write the number in the form $a + ib$.

19. $z_1 = 2\left(\cos\dfrac{\pi}{8} + i\sin\dfrac{\pi}{8}\right),\ z_2 = 4\left(\cos\dfrac{3\pi}{8} + i\sin\dfrac{3\pi}{8}\right)$

20. $z_1 = \sqrt{2}\left(\cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4}\right),\ z_2 = \sqrt{3}\left(\cos\dfrac{\pi}{12} + i\sin\dfrac{\pi}{12}\right)$

In Problems 21–24, write each complex number in polar form. Then use either (6) or (7) to obtain the polar form of the given number. Finally, write the polar form in the form $a + ib$.

21. $(3 - 3i)(5 + 5\sqrt{3}i)$

22. $(4 + 4i)(-1 + i)$

23. $\dfrac{-i}{1+i}$

24. $\dfrac{\sqrt{2} + \sqrt{6}i}{-1 + \sqrt{3}i}$

In Problems 25–30, use (9) to compute the indicated powers. Write the complex number in polar form and then in the form $a + ib$.

25. $(1 + \sqrt{3}i)^9$

26. $(2 - 2i)^5$

27. $\left(\frac{1}{2} + \frac{1}{2}i\right)^{10}$

28. $(-\sqrt{2} + \sqrt{6}i)^4$

29. $\left[\left(\sqrt{2}\cos\dfrac{\pi}{8} + i\sqrt{2}\sin\dfrac{\pi}{8}\right)\right]^{12}$

30. $\left[\sqrt{3}\left(\cos\dfrac{2\pi}{9} + i\sin\dfrac{2\pi}{9}\right)\right]^{6}$

In Problems 31 and 32, write the given complex number in polar form and then in the form $a + ib$.

31. $\left(\cos\frac{\pi}{9} + i\sin\frac{\pi}{9}\right)^{12}\left[2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^{5}$

32. $\dfrac{\left[8\left(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}\right)\right]^{3}}{\left[2\left(\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right)\right]^{10}}$

33. Use de Moivre's formula (10) with $n = 2$ to find trigonometric identities for $\cos 2\theta$ and $\sin 2\theta$.

34. Use de Moivre's formula (10) with $n = 3$ to find trigonometric identities for $\cos 3\theta$ and $\sin 3\theta$.

In Problems 35 and 36, find a positive integer n for which the equality holds.

35. $\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{n} = -1$ **36.** $\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)^{n} = 1$

37. For the complex numbers $z_1 = -1$ and $z_2 = 5i$, verify that:

(a) $\text{Arg}(z_1 z_2) \neq \text{Arg}(z_1) + \text{Arg}(z_2)$

(b) $\text{Arg}(-z_2/z_1) \neq \text{Arg}(-z_2) - \text{Arg}(z_1)$.

38. For the complex numbers given in Problem 37, verify that:

(a) $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

(b) $\arg(-z_2/z_1) = \arg(-z_2) - \arg(z_1)$.

Focus on Concepts

39. Suppose that $z = r(\cos\theta + i\sin\theta)$. Describe geometrically the effect of multiplying z by a complex number of the form $z_1 = \cos\alpha + i\sin\alpha$, when $\alpha > 0$ and when $\alpha < 0$.

40. Suppose $z = \cos\theta + i\sin\theta$. If n is an integer, evaluate $z^n + \bar{z}^n$ and $z^n - \bar{z}^n$.

41. Write an equation that relates $\arg(z)$ to $\arg(1/z)$, $z \neq 0$.

42. Are there any special cases in which $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$? Prove your assertions.

43. How are the complex numbers z_1 and z_2 related if $\arg(z_1) = \arg(z_2)$?

44. Describe the set of points z in the complex plane that satisfy $\arg(z) = \pi/4$.

45. Student A states that, even though she can't find it in the text, she thinks that $\arg(\bar{z}) = -\arg(z)$. For example, she says, if $z = 1 + i$, then $\bar{z} = 1 - i$ and $\arg(z) = \pi/4$ and $\arg(\bar{z}) = -\pi/4$. Student B disagrees because he feels that he has a counterexample: If $z = i$, then $\bar{z} = -i$; we can take $\arg(i) = \pi/2$ and $\arg(-i) = 3\pi/2$ and so $\arg(i) \neq -\arg(-i)$. Take sides and defend your position.

46. Suppose z_1, z_2, and $z_1 z_2$ are complex numbers in the first quadrant and that the points $z = 0$, $z = 1$, z_1, z_2, and $z_1 z_2$ are labeled O, A, B, C, and D, respectively. Study the formula in (6) and then discuss how the triangles OAB and OCD are related.

47. Suppose $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 = z_2$, then how are r_1 and r_2 related? How are θ_1 and θ_2 related?

48. Suppose z_1 is in the first quadrant. For each z_2, discuss the quadrant in which $z_1 z_2$ could be located.

(a) $z_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ **(b)** $z_2 = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$

(c) $z_2 = -i$ **(d)** $z_2 = -1$

49. (**a**) For $z \neq 1$, verify the identity

$$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z}.$$

(**b**) Use part (a) and appropriate results from this section to establish that

$$1 + \cos\theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin\left(n + \frac{1}{2}\right)\theta}{2\sin\frac{1}{2}\theta}$$

for $0 < \theta < 2\pi$. The foregoing result is known as **Lagrange's identity** and is useful in the theory of Fourier series.

50. Suppose z_1, z_2, z_3, and z_4 are four distinct complex numbers. Interpret geometrically:

$$\arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = \frac{\pi}{2}.$$

51. For n an even positive integer, establish the formula

$$\cos n\theta = \cos^n\theta - \binom{n}{2}\cos^{n-2}\theta\sin^2\theta + \binom{n}{4}\cos^{n-4}\theta\sin^4\theta + \cdots + (-1)^{n/2}\sin^n\theta$$

where $\binom{n}{k}$ is the **binomial coefficient** defined by

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}.$$

Computer Lab Assignments

In Problems 52–55, use a CAS to find the modulus and principal argument of the given complex number. In *Mathematica* use the commands **Abs** and **Arg**.

52. $2 - 2i$

53. $\dfrac{-6i}{1 + \sqrt{3}i}$

54. $3.2 - 7.1i$

55. $\dfrac{(1.4 + 2.0i)(0.3 - 1.2i)}{6.5 - 3.9i}$

1.4 Powers and Roots

INTRODUCTION Recall from algebra that -2 and 2 are said to be *square roots* of the number 4 because $(-2)^2 = 4$ and $(2)^2 = 4$. In other words, the two square roots of 4 are distinct solutions of the equation $w^2 = 4$. In like manner we say $w = 3$ is a cube root of 27 since $w^3 = 3^3 = 27$. This last equation points us again in the direction of complex variables since any real number has only *one real* cube root and *two complex* roots. In general, we say that a number w is an ***n*th root** of a nonzero complex number z if $w^n = z$, where n is a positive integer. For example, you are urged to verify that $w_1 = \frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}i$ and $w_2 = -\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}i$ are the two square roots of the complex number $z = i$ because $w_1^2 = i$ and $w_2^2 = i$. See Problem 39 in Exercises 1.1.

We will now demonstrate that there are exactly n solutions of the equation $w^n = z$.

Roots

Suppose $z = r(\cos\theta + i\sin\theta)$ and $w = \rho(\cos\phi + i\sin\phi)$ are polar forms of the complex numbers z and w. Then, in view of (9) of Section 1.3, the equation $w^n = z$ becomes

$$\rho^n(\cos n\phi + i\sin n\phi) = r(\cos\theta + i\sin\theta). \tag{1}$$

From (1), we can conclude that

$$\rho^n = r \tag{2}$$

and

$$\cos n\phi + i \sin n\phi = \cos\theta + i \sin\theta. \tag{3}$$

See Problem 47 in Exercises 1.3.

From (2), we define $\rho = \sqrt[n]{r}$ to be the unique positive nth root of the positive real number r. From (3), the definition of equality of two complex numbers implies that

$$\cos n\phi = \cos\theta \quad \text{and} \quad \sin n\phi = \sin\theta.$$

These equalities, in turn, indicate that the arguments θ and ϕ are related by $n\phi = \theta + 2k\pi$, where k is an integer. Thus,

$$\phi = \frac{\theta + 2k\pi}{n}.$$

As k takes on the successive integer values $k = 0,\ 1,\ 2,\ \ldots,\ n-1$ we obtain n *distinct* nth roots of z; these roots have the same modulus $\sqrt[n]{r}$ but different arguments. Notice that for $k \geq n$ we obtain the same roots because the sine and cosine are 2π-periodic. To see why this is so, suppose $k = n + m$, where $m = 0,\ 1,\ 2, \ldots$. Then

$$\phi = \frac{\theta + 2(n+m)\pi}{n} = \frac{\theta + 2m\pi}{n} + 2\pi$$

and

$$\sin\phi = \sin\left(\frac{\theta + 2m\pi}{n}\right), \quad \cos\phi = \cos\left(\frac{\theta + 2m\pi}{n}\right).$$

We summarize this result. The n nth roots of a nonzero complex number $z = r(\cos\theta + i\sin\theta)$ are given by

$$w_k = \sqrt[n]{r}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right], \tag{4}$$

where $k = 0,\ 1,\ 2,\ \ldots,\ n-1$.

EXAMPLE 1 Cube Roots of a Complex Number

Find the three cube roots of $z = i$.

Solution Keep in mind that we are basically solving the equation $w^3 = i$. Now with $r = 1$, $\theta = \arg(i) = \pi/2$, a polar form of the given number is given by $z = \cos(\pi/2) + i\sin(\pi/2)$. From (4), with $n = 3$, we then obtain

$$w_k = \sqrt[3]{1}\left[\cos\left(\frac{\pi/2 + 2k\pi}{3}\right) + i\sin\left(\frac{\pi/2 + 2k\pi}{3}\right)\right], \quad k = 0, 1, 2.$$

Hence the three roots are,

$$k = 0, \quad w_0 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 1, \quad w_1 = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 2, \quad w_2 = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} = -i.$$

Principal *n*th Root

On page 16 we pointed out that the symbol $\arg(z)$ really stands for a *set* of arguments for a complex number z. Stated another way, for a given complex number $z \neq 0$, $\arg(z)$ is *infinite-valued*. In like manner, $z^{1/n}$ is n-valued; that is, the symbol $z^{1/n}$ represents the set of n nth roots w_k of z. The *unique* root of a complex number z (obtained by using the principal value of $\arg(z)$ with $k = 0$) is naturally referred to as the **principal *n*th root** of w. In Example 1, since $\text{Arg}(i) = \pi/2$, we see that $w_0 = \frac{1}{2}\sqrt{3} + \frac{1}{2}i$ is the principal cube root of i. The choice of $\theta = \text{Arg}(z)$ and $k = 0$ guarantees us that when z is a positive real number r, the principal nth root is $\sqrt[n]{r}$.

$\sqrt{4} = 2$ and $\sqrt[3]{27} = 3$ are the principal square root of 4 and the principal cube root of 27, respectively. ▶

Since the roots given by (4) have the same modulus, the n nth roots of a nonzero complex number z lie on a circle of radius $\sqrt[n]{r}$ centered at the origin in the complex plane. Moreover, since the difference between the arguments of any two successive roots w_k and w_{k+1} is $2\pi/n$, the n nth roots of z are equally spaced on this circle, beginning with the root whose argument is θ/n. FIGURE 1.4.1 shows the three cube roots of i obtained in Example 1 spaced at equal angular intervals of $2\pi/3$ on the circumference of a unit circle beginning with the root w_0 whose argument is $\pi/6$.

As the next example shows, the roots of a complex number do not have to be "nice" numbers as in Example 1.

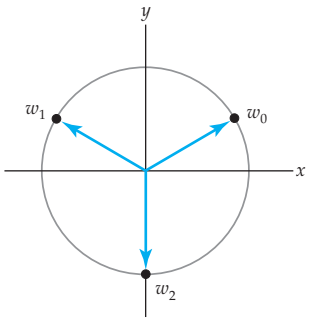


FIGURE 1.4.1 Three cube roots of i

EXAMPLE 2 Fourth Roots of a Complex Number

Find the four fourth roots of $z = 1 + i$.

Solution In this case, $r = \sqrt{2}$ and $\theta = \arg(z) = \pi/4$. From (4) with $n = 4$, we obtain

$$w_k = \sqrt[8]{2}\left[\cos\left(\frac{\pi/4 + 2k\pi}{4}\right) + i\sin\left(\frac{\pi/4 + 2k\pi}{4}\right)\right], \quad k = 0,\ 1,\ 2,\ 3.$$

With the aid of a calculator we find

$$\begin{aligned}
k = 0, \quad & w_0 = \sqrt[8]{2}\left[\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right] \approx 1.0696 + 0.2127i \\
k = 1, \quad & w_1 = \sqrt[8]{2}\left[\cos\frac{9\pi}{16} + i\sin\frac{9\pi}{16}\right] \approx -0.2127 + 1.0696i \\
k = 2, \quad & w_2 = \sqrt[8]{2}\left[\cos\frac{17\pi}{16} + i\sin\frac{17\pi}{16}\right] \approx -1.0696 - 0.2127i \\
k = 3, \quad & w_3 = \sqrt[8]{2}\left[\cos\frac{25\pi}{16} + i\sin\frac{25\pi}{16}\right] \approx 0.2127 - 1.0696i.
\end{aligned}$$

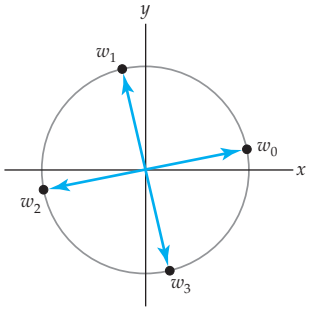


FIGURE 1.4.2 Four fourth roots of $1 + i$

As shown in FIGURE 1.4.2, the four roots lie on a circle centered at the origin of radius $r = \sqrt[8]{2} \approx 1.09$ and are spaced at equal angular intervals of $2\pi/4 = \pi/2$ radians, beginning with the root whose argument is $\pi/16$. Since $\theta = \text{Arg}(z) = \pi/4$, the principal fourth root is w_0. ☰

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) As a consequence of (4), we can say that the complex number system is *closed* under the operation of extracting roots. This means that for any z in **C**, $z^{1/n}$ is also in **C**. The real number system does not possess a similar closure property since, if x is in **R**, $x^{1/n}$ is not necessarily in **R**.

(*ii*) Geometrically, the n nth roots of a complex number z can also be interpreted as the vertices of a regular polygon with n sides that is inscribed within a circle of radius $\sqrt[n]{r}$ centered at the origin. You can see the plausibility of this fact by reinspecting Figures 1.4.1 and 1.4.2. See Problem 19 in Exercises 1.4.

(*iii*) When m and n are positive integers with no common factors, then (4) enables us to define a **rational power** of z, that is, $z^{m/n}$. It can be shown that the set of values $(z^{1/n})^m$ is the same as the set of values $(z^m)^{1/n}$. This set of n common values is defined to be $z^{m/n}$. See Problems 25 and 26 in Exercises 1.4.

EXERCISES 1.4

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–14, use (4) to compute all roots. Give the principal nth root in each case. Sketch the roots $w_0, w_1, \ldots, w_{n-1}$ on an appropriate circle centered at the origin.

1. $(8)^{1/3}$

2. $(-1)^{1/4}$

3. $(-9)^{1/2}$

4. $(-125)^{1/3}$

5. $(i)^{1/2}$

6. $(-i)^{1/3}$

7. $(-1+i)^{1/3}$

8. $(1+i)^{1/5}$

9. $(-1+\sqrt{3}i)^{1/2}$

10. $(-1-\sqrt{3}i)^{1/4}$

11. $(3+4i)^{1/2}$

12. $(5+12i)^{1/2}$

13. $\left(\dfrac{16i}{1+i}\right)^{1/8}$

14. $\left(\dfrac{1+i}{\sqrt{3}+i}\right)^{1/6}$

15. (**a**) Verify that $(4+3i)^2 = 7+24i$.

(**b**) Use part (a) to find the two values of $(7+24i)^{1/2}$.

16. Rework Problem 15 using (4).

17. Find all solutions of the equation $z^4 + 1 = 0$.

18. Use the fact that $8i = (2+2i)^2$ to find all solutions of the equation

$$z^2 - 8z + 16 = 8i.$$

The n distinct ***n*th roots of unity** are the solutions of the equation $w^n = 1$. Problems 19–24 deal with roots of unity.

19. (**a**) Show that the n nth roots of unity are given by

$$(1)^{1/n} = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}, \quad k = 0, 1, 2, \ldots, n-1.$$

(**b**) Find n nth roots of unity for $n = 3$, $n = 4$, and $n = 5$.

(**c**) Carefully plot the roots of unity found in part (b). Sketch the regular polygons formed with the roots as vertices. [*Hint*: See (*ii*) in the *Remarks*.]

20. Suppose w is a cube root of unity corresponding to $k = 1$. See Problem 19(a).

(**a**) How are w and w^2 related?

(**b**) Verify by direct computation that

$$1 + w + w^2 = 0.$$

(**c**) Explain how the result in part (b) follows from the basic definition that w is a cube root of 1, that is, $w^3 = 1$. [*Hint*: Factor.]

21. For a fixed n, if we take $k = 1$ in Problem 19(a), we obtain the root

$$w_n = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}.$$

Explain why the n nth roots of unity can then be written

$$1, w_n, w_n^2, w_n^3, \ldots, w_n^{n-1}.$$

22. Consider the equation $(z + 2)^n + z^n = 0$, where n is a positive integer. By any means, solve the equation for z when $n = 1$. When $n = 2$.

23. Consider the equation in Problem 22.

(**a**) In the complex plane, determine the location of all solutions z when $n = 5$. [*Hint*: Write the equation in the form $[(z + 2)/(-z)]^5 = 1$ and use part (a) of Problem 19.]

(**b**) Reexamine the solutions of the equation in Problem 22 for $n = 1$ and $n = 2$. Form a conjecture as to the location of all solutions of $(z + 2)^n + z^n = 0$.

24. For the n nth roots of unity given in Problem 21, show that

$$1 + w_n + w_n^2 + w_n^3 + \cdots + w_n^{n-1} = 0.$$

[*Hint*: Multiply the sum $1 + w_n + w_n^2 + w_n^3 + \cdots + w_n^{n-1}$ by $w_n - 1$.]

Before working Problems 25 and 26, read (*iii*) in the *Remarks*. If m and n are positive integers with no common factors, then the n values of the rational power $z^{m/n}$ are

$$w_k = \sqrt[n]{r^m}\left[\cos\frac{m}{n}(\theta + 2k\pi) + i\sin\frac{m}{n}(\theta + 2k\pi)\right], \tag{5}$$

$k = 0,\ 1,\ 2,\ \ldots,\ n - 1$. The w_k are the n distinct solutions of $w^n = z^m$.

25. (**a**) First compute the set of values $i^{1/2}$ using (4). Then compute $(i^{1/2})^3$ using (9) of Section 1.3.

(**b**) Now compute i^3. Then compute $(i^3)^{1/2}$ using (4). Compare these values with the results of part (b).

(**c**) Lastly, compute $i^{3/2}$ using formula (5).

26. Use (5) to find all solutions of the equation $w^2 = (-1 + i)^5$.

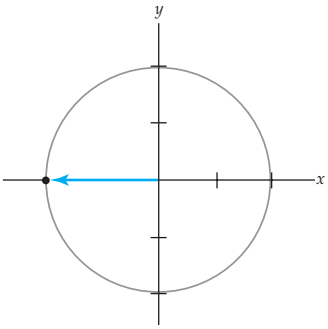


FIGURE 1.4.3 Figure for Problem 27

Focus on Concepts

27. The vector given in **FIGURE 1.4.3** represents one value of $z^{1/n}$. Using only the figure and trigonometry—that is, do not use formula (4)—find the remaining values of $z^{1/n}$ when $n = 3$. Write the values of $z^{1/n}$ in the form $a + ib$. Repeat for $n = 4$.

28. Suppose n denotes a nonnegative integer. Determine the values of n such that $z^n = 1$ possesses only real solutions. Defend your answer with sound mathematics.

29. (**a**) Proceed as in Example 2 to find the approximate values of the two square roots w_0 and w_1 of $1 + i$.

(**b**) Show that the exact values of the roots in part (a) are

$$w_0 = \sqrt{\frac{1+\sqrt{2}}{2}} + i\sqrt{\frac{\sqrt{2}-1}{2}}, \quad w_1 = -\sqrt{\frac{1+\sqrt{2}}{2}} - i\sqrt{\frac{\sqrt{2}-1}{2}}.$$

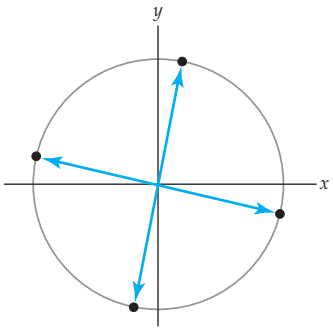


FIGURE 1.4.4 Figure for Problem 34

30. Discuss: What geometric significance does the result in Problem 24 have?

31. Discuss: A real number can have a complex nth root. Can a nonreal complex number have a real nth root?

32. Suppose w is located in the first quadrant and is a cube root of a complex number z. Can there exist a second cube root of z located in the first quadrant? Defend your answer with sound mathematics.

33. Suppose z is a complex number that possesses a fourth root w that is neither real nor pure imaginary. Explain why the remaining fourth roots are neither real nor pure imaginary.

34. Discuss: The vectors in FIGURE 1.4.4 represent the four fourth roots of a complex number z. Can you determine from this figure alone which root is the principal fourth root of z?

Computer Lab Assignments

In Problems 35–40, use a CAS to first find $z^n = w$ for the given complex number and the indicated value of n. Then, using the output and the same value of n, determine whether $w^{1/n} = (z^n)^{1/n} = z$. If not, explain why not.

35. $z = 2.5 - i;\ n = 10$

36. $z = -0.5 + 0.3i;\ n = 5$

37. $z = 1 + 3i;\ n = 8$

38. $z = 2 + 2i;\ n = 12$

39. $z = i;\ n = 21$

40. $z = -1 + \sqrt{3}\,i;\ n = 11$

1.5 Sets of Points in the Complex Plane

INTRODUCTION In the preceding sections we examined some rudiments of the algebra and geometry of complex numbers. But we have barely scratched the surface of the subject known as complex analysis; the main thrust of our study lies ahead. Our goal in the chapters that follow is to examine functions of a single complex variable $z = x + iy$ and the calculus of these functions.

Before introducing the notion of a function in Chapter 2, we need to state some essential definitions and terminology about sets in the complex plane.

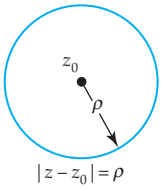


FIGURE 1.5.1 Circle of radius ρ

Circles Suppose $z_0 = x_0 + iy_0$. Since $|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2}$ is the distance between the points $z = x + iy$ and $z_0 = x_0 + iy_0$, the points $z = x + iy$ that satisfy the equation

$$|z - z_0| = \rho, \quad \rho > 0, \tag{1}$$

lie on a **circle** of radius ρ centered at the point z_0. See FIGURE 1.5.1.

EXAMPLE 1 Two Circles

(a) $|z| = 1$ is an equation of a unit circle centered at the origin.

(b) By rewriting $|z - 1 + 3i| = 5$ as $|z - (1 - 3i)| = 5$, we see from (1) that the equation describes a circle of radius 5 centered at the point $z_0 = 1 - 3i$. ☰

Disks and Neighborhoods The points z that satisfy the inequality $|z - z_0| \leq \rho$ can be either *on* the circle $|z - z_0| = \rho$ or *within* the circle. We say that the set of points defined by $|z - z_0| \leq \rho$ is a **disk** of radius ρ centered at z_0. But the points z that satisfy the strict inequality $|z - z_0| < \rho$ lie within, and not on, a circle of radius ρ centered at the point z_0. This set is called a **neighborhood** of z_0. Occasionally, we will need to use a neighborhood of z_0 that also *excludes* z_0. Such a neighborhood is defined by the simultaneous inequality $0 < |z - z_0| < \rho$ and is called a **deleted neighborhood** of z_0. For example, $|z| < 1$ defines a neighborhood of the origin, whereas $0 < |z| < 1$ defines a deleted neighborhood of the origin; $|z - 3 + 4i| < 0.01$ defines a neighborhood of $3 - 4i$, whereas the inequality $0 < |z - 3 + 4i| < 0.01$ defines a deleted neighborhood of $3 - 4i$.

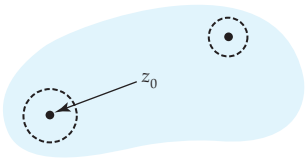


FIGURE 1.5.2 Open set

Open Sets A point z_0 is said to be an **interior point** of a set S of the complex plane if there exists some neighborhood of z_0 that lies entirely within S. If every point z of a set S is an interior point, then S is said to be an **open set**. See FIGURE 1.5.2. For example, the inequality $\text{Re}(z) > 1$ defines a *right half-plane*, which is an open set. All complex numbers $z = x + iy$ for which $x > 1$ are in this set. If we choose, for example, $z_0 = 1.1 + 2i$, then a neighborhood of z_0 lying entirely in the set is defined by $|z - (1.1 + 2i)| < 0.05$. See FIGURE 1.5.3. On the other hand, the set S of points in the complex plane defined by $\text{Re}(z) \geq 1$ is *not* open because every neighborhood of a point lying on the line $x = 1$ must contain points in S and points not in S. See FIGURE 1.5.4.

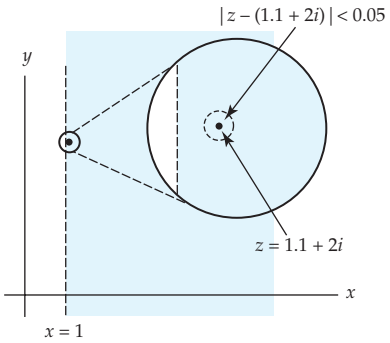


FIGURE 1.5.3 Open set with magnified view of point near $x = 1$

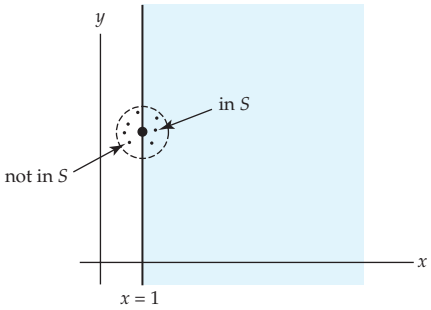
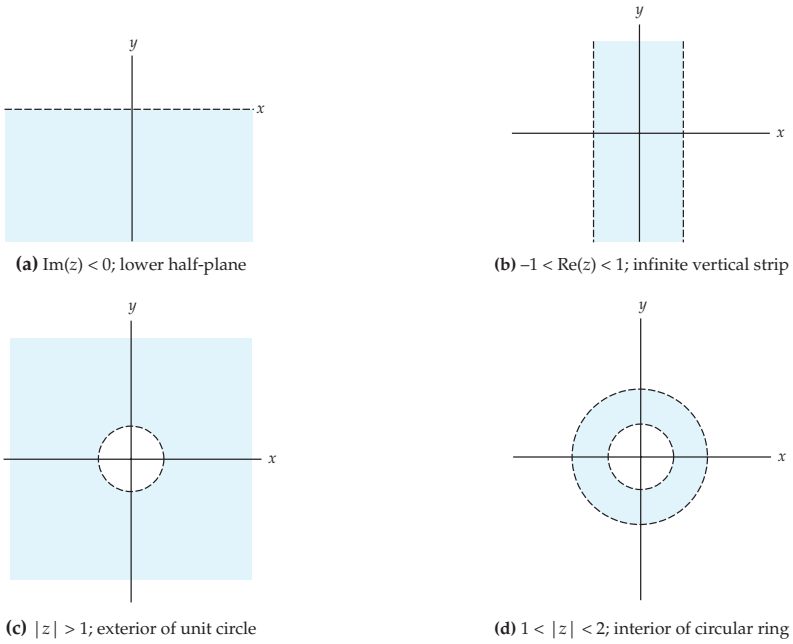


FIGURE 1.5.4 Set S not open

EXAMPLE 2 Some Open Sets

FIGURE 1.5.5 illustrates some additional open sets. ☰

If *every* neighborhood of a point z_0 of a set S contains at least one point of S and at least one point not in S, then z_0 is said to be a **boundary point** of S. For the set of points defined by $\text{Re}(z) \geq 1$, the points on the vertical line $x = 1$ are boundary



(a) Im(z) < 0; lower half-plane

(b) –1 < Re(z) < 1; infinite vertical strip

(c) |z| > 1; exterior of unit circle

(d) 1 < |z| < 2; interior of circular ring

FIGURE 1.5.5 Four examples of open sets

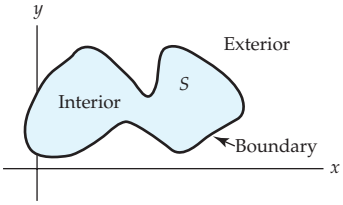


FIGURE 1.5.6 Interior, boundary, and exterior of set S

points. The points that lie on the circle $|z - i| = 2$ are boundary points for the disk $|z - i| \leq 2$ as well as for the neighborhood $|z - i| < 2$ of $z = i$. The collection of boundary points of a set S is called the **boundary** of S. The circle $|z - i| = 2$ is the boundary for both the disk $|z - i| \leq 2$ and the neighborhood $|z - i| < 2$ of $z = i$. A point z that is neither an interior point nor a boundary point of a set S is said to be an **exterior point** of S; in other words, z_0 is an exterior point of a set S if there exists *some* neighborhood of z_0 that contains no points of S. FIGURE 1.5.6 shows a typical set S with interior, boundary, and exterior.

An open set S can be as simple as the complex plane with a single point z_0 deleted. The boundary of this "punctured plane" is z_0, and the only candidate for an exterior point is z_0. However, S has no exterior points since no neighborhood of z_0 can be free of points of the plane.

Annulus The set S_1 of points satisfying the inequality $\rho_1 < |z - z_0|$ lie exterior to the circle of radius ρ_1 centered at z_0, whereas the set S_2 of points satisfying $|z - z_0| < \rho_2$ lie interior to the circle of radius ρ_2 centered at z_0. Thus, if $0 < \rho_1 < \rho_2$, then the set of points satisfying the simultaneous inequality

$$\rho_1 < |z - z_0| < \rho_2, \tag{2}$$

is the intersection of the sets S_1 and S_2. This intersection is an open circular ring centered at z_0. Figure 1.5.5(d) illustrates such a ring centered at the origin. The set defined by (2) is called an open **circular annulus**. By allowing $\rho_1 = 0$, we obtain a deleted neighborhood of z_0.

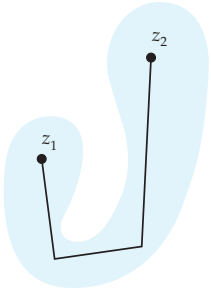


FIGURE 1.5.7 Connected set

Domain If any pair of points z_1 and z_2 in a set S can be connected by a polygonal line that consists of a finite number of line segments joined end to end that lies entirely in the set, then the set S is said to be **connected**. See FIGURE 1.5.7. An open connected set is called a **domain**. Each of the open sets in Figure 1.5.5 is connected and so each is a domain. The set of numbers z satisfying $\text{Re}(z) \neq 4$ is an open set but is not connected since it is not possible to join points on either side of

the vertical line $x = 4$ by a polygonal line without leaving the set (bear in mind that the points on the line $x = 4$ are not in the set). A neighborhood of a point z_0 is a connected set.

Regions A **region** is a domain in the complex plane together with all, some, or none of its boundary points. Since an open connected set does not contain any boundary points, it is automatically a region. A set that contains all of its boundary points is said to be **closed**. The disk defined by $|z - z_0| \leq \rho$ is an example of a closed region and is referred to as a **closed disk**. A neighborhood of a point z_0 defined by $|z - z_0| < \rho$ is an open set or an open region and is said to be an **open disk**. If the center z_0 is deleted from either a closed disk or an open disk, the regions defined by $0 < |z - z_0| \leq \rho$ or $0 < |z - z_0| < \rho$ are called **punctured disks**. *A punctured open disk is the same as a deleted neighborhood of* z_0. A region can be neither open nor closed; the annular region defined by the inequality $1 \leq |z - 5| < 3$ contains only some of its boundary points (the points lying on the circle $|z - 5| = 1$), and so it is *neither open nor closed*. In (2) we defined a circular annular region; in a more general interpretation, an annulus or **annular region** may have the appearance shown in FIGURE 1.5.8.

Note: "Closed" does not mean "not open." ▶

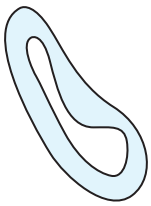


FIGURE 1.5.8 Annular region

Bounded Sets Finally, we say that a set S in the complex plane is **bounded** if there exists a real number $R > 0$ such that $|z| < R$ for every z in S. That is, S is bounded if it can be completely enclosed within *some* neighborhood of the origin. In FIGURE 1.5.9, the set S shown in color is bounded because it is contained entirely within the dashed circular neighborhood of the origin. A set is **unbounded** if it is not bounded. For example, the set in Figure 1.5.5(d) is bounded, whereas the sets in Figure 1.5.5(a), 1.5.5(b), and 1.5.5(c) are unbounded.

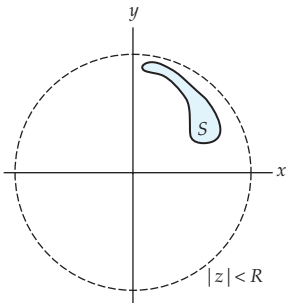


FIGURE 1.5.9 The set S is bounded since some neighborhood of the origin encloses S entirely

Remarks COMPARISON WITH REAL ANALYSIS

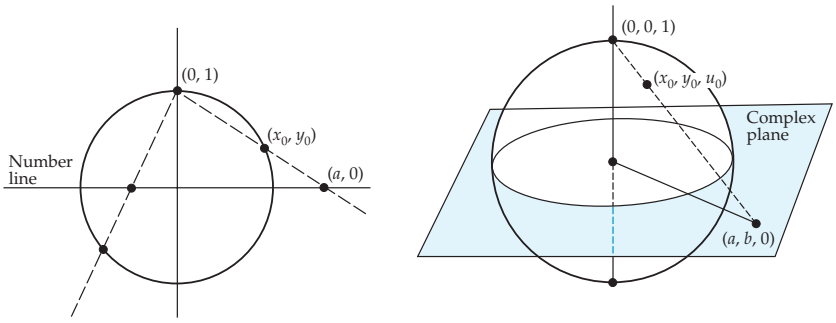
In your study of mathematics you undoubtedly found that you had to deal with the concept of *infinity*. For example, in a course in calculus you dealt with *limits at infinity*, where the behavior of functions was examined as x either increased or decreased without bound. Since we have exactly two directions on a number line, it is convenient to represent the notions of "increasing without bound" and "decreasing without bound" symbolically by $x \to +\infty$ and $x \to -\infty$, respectively. It turns out that we can get along just fine without the designation $\pm\infty$ by dealing with an "ideal point" called the **point at infinity,** which is denoted simply by ∞. To do this we identify any real number a with a point (x_0, y_0) on a unit circle $x^2 + y^2 = 1$ by drawing a straight line from the point $(a, 0)$ on the x-axis or horizontal number line to the point $(0, 1)$ on the circle. The point (x_0, y_0) is the point of intersection of the line with the circle. See Problem 47 in Exercises 1.5. It should be clear from FIGURE 1.5.10(a) that the farther the point $(a, 0)$ is from the origin, the nearer (x_0, y_0) is to $(0, 1)$. The only point on the circle that does not correspond to a real number a is $(0, 1)$. To complete the correspondence with *all* the points on the circle, we identify $(0, 1)$ with ∞. The set consisting of the real numbers $\mathbf{R}$ adjoined with ∞ is called the **extended real-number system**.

In our current study the analogue of the number line is the complex plane. Recall that since $\mathbf{C}$ is not ordered, the notions of z either "increasing" or "decreasing" have no meaning. However, we know that by increasing the modulus $|z|$ of a complex number z, the number moves farther from the origin. If we

allow $|z|$ to become unbounded, say, along the real and imaginary axes, we do not have to distinguish between "directions" on these axes by notations such as $z \to +\infty$, $z \to -\infty$, $z \to +i\infty$, and $z \to -i\infty$. In complex analysis, only a single notion of ∞ is used because we can extend the complex number system **C** in a manner analogous to that just described for the real number system **R**. This time, however, we associate a complex number with a point on a unit sphere called the **Riemann sphere**. By drawing a line from the number $z = a + ib$, written as $(a,\ b,\ 0)$, in the complex plane to the north pole $(0,\ 0,\ 1)$ of the sphere $x^2 + y^2 + u^2 = 1$, we determine a unique point $(x_0,\ y_0,\ u_0)$ on a unit sphere. As can be visualized from Figure 1.5.10(b), a complex number with a very large modulus is far from the origin (0, 0, 0) and, correspondingly, the point $(x_0,\ y_0,\ u_0)$ is close to $(0,\ 0,\ 1)$. In this manner each complex number is identified with a single point on the sphere. See Problems 48–50 in Exercises 1.5. Because the point $(0,\ 0,\ 1)$ corresponds to no number z in the plane, we correspond it with ∞. Of course, the system consisting of **C** adjoined with the "ideal point" ∞ is called the **extended complex-number system**.

This way of corresponding or mapping the complex numbers onto a sphere—north pole $(0,\ 0,\ 1)$ excluded—is called a **stereographic projection**.

For a finite number z, we have $z + \infty = \infty + z = \infty$, and for $z \neq 0$, $z \cdot \infty = \infty \cdot z = \infty$. Moreover, for $z \neq 0$ we write $z/0 = \infty$ and for $z \neq \infty$, $z/\infty = 0$. Expressions such as $\infty - \infty$, ∞/∞, ∞^0, and 1^∞ cannot be given a meaningful definition and are called indeterminate.



(a) Unit circle **(b)** Unit sphere

FIGURE 1.5.10 The method of correspondence in (b) is a stereographic projection

EXERCISES 1.5

Answers to selected odd-numbered problems begin on page ANS-3.

In Problems 1–12, sketch the graph of the given equation in the complex plane.

1. $|z - 4 + 3i| = 5$

2. $|z + 2 + 2i| = 2$

3. $|z + 3i| = 2$

4. $|2z - 1| = 4$

5. $\text{Re}(z) = 5$

6. $\text{Im}(z) = -2$

7. $\text{Im}(\bar{z} + 3i) = 6$

8. $\text{Im}(z - i) = \text{Re}(z + 4 - 3i)$

9. $|\text{Re}\,(1 + i\bar{z})| = 3$

10. $z^2 + \bar{z}^2 = 2$

11. $\text{Re}(z^2) = 1$

12. $\arg(z) = \pi/4$

In Problems 13–24, sketch the set S of points in the complex plane satisfying the given inequality. Determine whether the set is (**a**) open, (**b**) closed, (**c**) a domain, (**d**) bounded, or (**e**) connected.

13. $\text{Re}(z) < -1$

14. $|\text{Re}\,(z)| > 2$

15. $\text{Im}(z) > 3$

16. $\text{Re}\,((2+i)z+1) > 0$

17. $2 < \text{Re}(z-1) < 4$

18. $-1 \le \text{Im}(z) < 4$

19. $\text{Re}(z^2) > 0$

20. $\text{Im}(z) < \text{Re}(z)$

21. $|z-i| > 1$

22. $2 < |z-i| < 3$

23. $1 \le |z-1-i| < 2$

24. $2 \le |z-3+4i| \le 5$

25. Give the boundary points of the sets in Problems 13–24.

26. Consider the set S consisting of the complex plane with the circle $|z| = 5$ deleted. Give the boundary points of S. Is S connected?

In Problems 27 and 28, sketch the set of points in the complex plane satisfying the given inequality.

27. $0 \le \arg(z) \le \pi/6$

28. $-\pi < \arg(z) < \pi/2$

In Problems 29 and 30, describe the shaded set in the given figure using $\arg(z)$ and an inequality.

29.

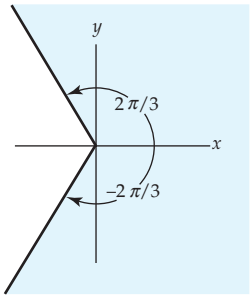


FIGURE 1.5.11 Figure for Problem 29

30.

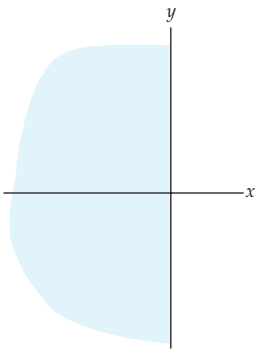


FIGURE 1.5.12 Figure for Problem 30

In Problems 31 and 32, solve the given pair of simultaneous equations.

31. $|z| = 2, \quad |z-2| = 2$

32. $|z-i| = 5, \quad \arg(z) = \pi/4$

Focus on Concepts

33. On page 29 we stated that if $\rho_1 > 0$, then the set of points satisfying $\rho_1 < |z - z_0|$ is the exterior to the circle of radius ρ_1 centered at z_0. In general, describe the set if $\rho_1 = 0$. In particular, describe the set defined by $|z + 2 - 5i| > 0$.

34. (**a**) What are the boundary points of a deleted neighborhood of z_0?

(**b**) What are the boundary points of the complex plane?

(**c**) Give several examples, not including the one given on page 30, of a set S in the complex plane that is neither open nor closed.

35. Use complex notation and inequalities in parts (a) and (b).

(**a**) Make up a list of five sets in the complex plane that are connected.

(**b**) Make up a list of five sets in the complex plane that are not connected.

36. Consider the disk centered at z_0 defined by $|z - z_0| \leq \rho$. Demonstrate that this set is bounded by finding an $R > 0$ so that all points z in the disk satisfy $|z| < R$. [*Hint*: See the discussion on inequalities in Section 1.2.]

37. Suppose z_0 and z_1 are distinct points. Using only the concept of distance, describe in words the set of points z in the complex plane that satisfy $|z - z_0| = |z - z_1|$.

38. Using only the concept of distance, describe in words the set of points z in the complex plane that satisfies $|z - i| + |z + i| = 1$.

In Problems 39 and 40, describe the shaded set in the given figure by filling in the two blanks in the set notation

$$\{\, z\text{: } _____ \text{ and/or } _____ \}$$

using complex notation for equations or inequalities and one of the words *and* or *or*.

39.

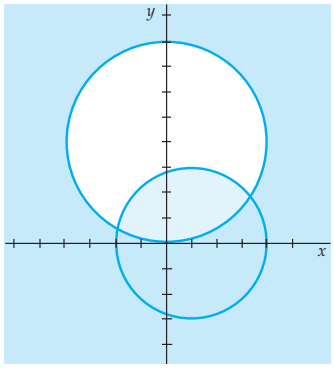


FIGURE 1.5.13 Figure for Problem 39

40.

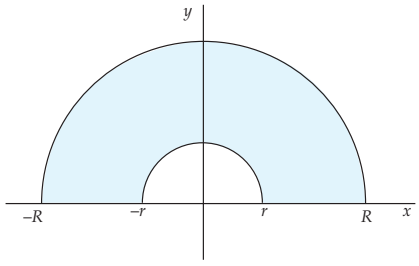


FIGURE 1.5.14 Figure for Problem 40

41. Consider the set S of points in the complex plane defined by $\{i/n\}$, $n = 1, 2, 3, \ldots$. Discuss which of the following terms apply to S: *boundary*, *open*, *closed*, *connected*, *bounded*.

42. Consider a finite set S of complex numbers $\{z_1, z_2, z_3, \ldots, z_n\}$. Discuss whether S is necessarily bounded. Defend your answer with sound mathematics.

43. A set S is said to be **convex** if each pair of points P and Q in S can be joined by a line segment $\overline{PQ}$ such that every point on the line segment also lies in S. Determine which of the sets S in the complex plane defined by the following conditions are convex.

(a) $|z - 2 + i| < 3$

(b) $1 < |z| < 2$

(c) $x > 2,\ y \leq -1$

(d) $y < x^2$

(e) $\text{Re}(z) \leq 5$

(f) $\text{Re}(z) \neq 0$

44. Discuss: Is a convex set, defined in Problem 43, necessarily connected?

45. Discuss and decide whether the empty set $\varnothing$ is an open set.

46. Suppose S_1 and S_2 are open sets in the complex plane.

(a) Discuss: Is the union $S_1 \cup S_2$ an open set? If you think the statement is true, try to prove it. If you think the statement is false, find a counterexample.

(b) Repeat part (a) for the intersection $S_1 \cap S_2$.

Before answering Problems 47–50, reread the *Remarks* on the end of this section.

47. Find the point $(x_0,\ y_0)$ on the unit circle that corresponds to each of the real numbers: $-\frac{1}{4},\ \frac{1}{2},\ -3,\ 1,\ 10$. See Figure 1.5.10(a).